CSV File

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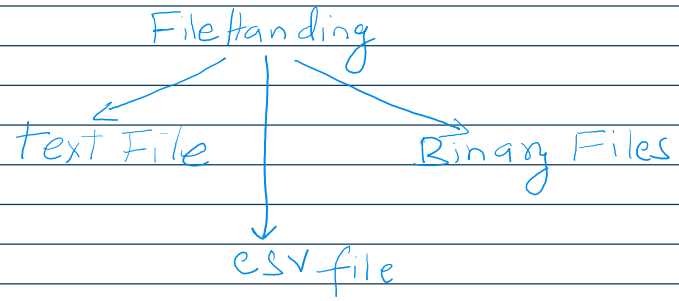

What is CSV File?

A CSV (comma-separated values) File is a plain text file that contains tabular data. Each line is a record and each field is separated by comma(,).

FileHandling > Files > person.csv

```
Name,Age,City,Email,MobileNumber
Alice,25,New York,aliceh8@gmail.com,7862192121
Bob,30,London,bobm@gmail.com,3456781239
Charlie,22,Mumbai,charlieJ@outlook.com,2345678765
```

CSV File Handling in Python

Python provides two main ways to work with CSV files.

1. Using the built in CSV module.

2. Using the pandas library. (more powerful and easier for large data).

Using the csv module to read a csv file

```python
import csv
filename = input('Enter file name with .csv extention: ')
with open('./FileHandling/Files/'+filename) as fileObject:
    reader = csv.reader(fileObject)
    for row in reader:
        print (row)
```

Output:

```
Enter file name with .csv extention: person.csv
['Name', 'Age', 'City', 'Email', 'MobileNumber']
['Alice', '25', 'New York', 'aliceh8@gmail.com', '7862192121']
['Bob', '30', 'London', 'bobm@gmail.com', '3456781239']
['Charlie', '22', 'Mumbai', 'charlieJ@outlook.com', '2345678765']
```

Writing to a csv file

Code:

```python
import csv
fileName = input('Enter the file name with .csv extension: ')
with open('./FileHandling/Files/'+fileName, 'a', newline='') as fileObject:
    writer = csv.writer(fileObject)
```

```
    while True:
        name = input('Enter your name: ')
        age = input('Enter your age: ')
        city = input('Enter your city: ')
        email = input('Enter email: ')
        mobileNum = input('Enter mobile nummber: ')

        data = [name, age, city, email, mobileNum]
        writer.writerow(data)
        ch = input("Press 'y' to enter another record, other wise press 'n': ")
        if ch.lower() == 'n':
            print('Insertions are done.')
            break
```

Output:

```
Enter the file name with .csv extension: person.csv
Enter your name: Jiya
Enter your age: 11
Enter your city: Bengaluru
Enter email: jiyaG@gmail.com
Enter mobile nummber: 9876567890
Press 'y' to enter another record, other wise press 'n': y
Enter your name: Aadya
Enter your age: 17
Enter your city: Dehradun
Enter email: aadyaS@gmail.com
Enter mobile nummber: 1234509876
Press 'y' to enter another record, other wise press 'n': n
Insertions are done.
```

Reading with DictReader
This reads row as dictionaries using header as keys:

```
import csv
fileName = input('Enter file name: ')
with open('./FileHandling/Files/'+fileName, 'r') as fo:
    reader = csv.DictReader(fo)

    for row in reader:
        print(row['Name'], row['City'], row['Age'], row['MobileNumber'], row['Email'])
```

Output:

```
Enter file name: person.csv
Alice New York 25 7862192121 aliceh8@gmail.com
Bob London 30 3456781239 bobm@gmail.com
Charlie Mumbai 22 2345678765 charlieJ@outlook.com
Jiya Bengaluru 11 9876567890 jiyaG@gmail.com
Aadya Dehradun 17 1234509876 aadyaS@gmail.com
```

Writing with DictWriter

```python
import csv
fileName = input("Enter file name with .csv extention: ")
with open('./FileHandling/Files/' + fileName, 'a', newline='') as fo:
    fieldNames = ['Name', 'Age', 'City', 'Email', 'MobileNumber']
    writer = csv.DictWriter(fo, fieldNames)

    while True:
        name = input('Enter your name: ')
        age = input('Enter your age: ')
        city = input('Enter your city: ')
        email = input('Enter email: ')
        mobileNum = input('Enter mobile nummber: ')

        data = {'Name':name, 'Age':age, 'City':city, 'Email': email, 'MobileNumber':
mobileNum}
        writer.writerow(data)
        ch = input("Press 'y' to enter another record, other wise press 'n': ")
        if ch.lower() == 'n':
            print('Insertions are done.')
            break
```

Output:

```
Enter file name with .csv extention: person.csv
Enter your name: Ian
Enter your age: 21
Enter your city: Kolkata
Enter email: ian@gmail.com
Enter mobile nummber: 5678998765
Press 'y' to enter another record, other wise press 'n': n
Insertions are done.
```

Person.csv File:

FileHandling > Files > person.csv

```
Name,Age,City,Email,MobileNumber
Alice,25,New York,aliceh8@gmail.com,7862192121
Bob,30,London,bobm@gmail.com,3456781239
Charlie,22,Mumbai,charlieJ@outlook.com,2345678765
Jiya,11,Bengaluru,jiyaG@gmail.com,9876567890
Aadya,17,Dehradun,aadyaS@gmail.com,1234509876
Ian,21,Kolkata,ian@gmail.com,5678998765
```

Common parameters in csv.reader and writer

Parameter	Description
delimiter	Character used to separate fields.
quotechar	Used to quote fields with special character.
lineterminator	Sets line break character

Example:

```
    reader = csv.reader(fileObject, delimiter=';', quotechar='"', newline='')
```